

# Hints, Placebos and Shortcuts: Controlled Tests of Structural Primers for LLM Graph Reasoning

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## Abstract

Prepending a *primer* of factual sentences about a graph’s structure (node degree, clustering coefficient, random-walk return probabilities or component count) to its text encoding is a natural way to assist a language model on graph-reasoning questions. An apparent gain is uninformative if the answer can be read from the primer without consulting the graph, and confounded if the primer merely lengthens the prompt. We evaluate primers against four controls jointly: a graph-blind shortcut solver refit at the evaluated graph size, a length-matched content-free filler, a minimal-length content control, and a degree-preserving rewiring that varies triangle count at fixed prompt length. Across four Qwen3 arms (1.7B and 4B, with and without thinking mode) on 40-node Erdős–Rényi graphs, primers that state the answer help a model at floor and harm a model at ceiling (cross-fitted interaction with baseline accuracy  $-11.0$  points, 95% CI  $[-17.1, -5.7]$ ), whereas primers that do not state it show no such relation. The pattern replicates on eleven held-out arms (0.6–14B, two families) but is not graded at intermediate baselines. The only effect of a primer that does not state the answer, `clustering on node_degree`, is small relative to no primer (+1.9 points) and confined to intermediate densities; two candidate mechanisms for it are not supported.

## 1 Introduction

Fatemi et al. (2023) show that a large language model’s accuracy on graph-reasoning questions varies substantially with the choice of text encoding when the graph and question are held fixed. We ask a complementary question about the same benchmark: does prepending a short *primer* of derived, node-level facts to the graph encoding improve the model’s answers?

Four confounds complicate the interpretation of such an effect. First, a primer may help because

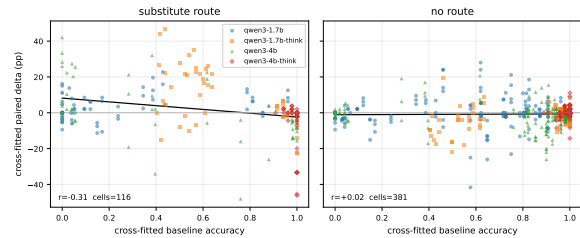


Figure 1: Cross-fitted paired effect against cross-fitted baseline accuracy (Section 5.4) for the 116 cells in which a primer offers a substitute route to the answer (left) and the 381 that do not (right). Each cell appears once per fold direction. Because baseline and effect are estimated from disjoint halves of the paired graphs, shared sampling noise cannot produce the slope; the bounded accuracy scale can, which Section 5.4 examines separately.

the model reads the answer from it rather than reasoning over the graph (answer leakage). Second, a primer changes prompt length, and context length and position alone affect retrieval and reasoning accuracy (Liu et al., 2024; Kamradt, 2023; Levy et al., 2024). Third, answer extraction can err in ways that correlate with response style, which appears as a spurious condition effect (Section 3.4). Fourth, a model near ceiling on a task leaves no room for any primer effect to appear.

We address three research questions. **RQ1:** when a primer improves accuracy, is the improvement attributable to the primer stating the answer? **RQ2:** does added text affect accuracy independently of its content? **RQ3:** does any primer that does not state the answer improve accuracy beyond a length-matched control, and through what mechanism? Our contributions are (i) a battery of controls applied jointly: a graph-blind shortcut solver refit at the evaluated graph size (Section 3.2), a length-matched content-free filler, a minimal-length content control (`components`), and a degree-preserving rewiring that changes triangle count at fixed prompt length (Section 3.3);

(ii) an empirical map of primer effects on 40-node Erdős–Rényi graphs across four model arms and six tasks; (iii) evidence that answer-stating primers act as a substitute route whose value depends on the model’s own baseline accuracy, supported by a cross-fitted interaction test and replicated on held-out models and under node renaming; and (iv) a characterisation of the single effect of a primer that does not state the answer, together with two mechanism tests that it fails.

## 2 Related Work

We adopt the GraphQA task definitions of [Fatemi et al. \(2023\)](#) and manipulate graph size and edge density directly, so that difficulty is an experimental variable rather than an incidental one. [Wang et al. \(2023\)](#) report that LLMs are brittle to spurious correlations in graph problems and propose *Build-a-Graph* and *Algorithmic* prompting, the prompt-level interventions closest to a primer; [Perozzi et al. \(2024\)](#) instead learn a continuous graph encoding. [Guo et al. \(2023\)](#) find that models struggle to exploit graph description languages directly, and [Jin et al. \(2024\)](#) survey the wider literature. [Dziri et al. \(2023\)](#) argue that transformers solve compositional tasks by linearised subgraph matching rather than systematic reasoning, which raises the same question as a primer’s benefit: does it reflect reasoning or a shorter pattern to match? [Geirhos et al. \(2020\)](#) frame this as *shortcut learning*; our shortcut solver operationalises that concern for the primer channel. Irrelevant context degrades reasoning ([Shi et al., 2023](#)), as does input length well below the context limit ([Levy et al., 2024](#)); our *filler* condition measures this cost directly. The *rwse* condition adapts the random-walk structural encoding of [Dwivedi et al. \(2022\)](#), rendered as sentences, and the rewiring follows the double-edge-swap configuration model ([Fosdick et al., 2018](#)). For statistics we follow [Dror et al. \(2018\)](#) on paired testing, test differences between groups directly rather than contrasting their significance ([Gelman and Stern, 2006](#)), and borrow cross-fitting from [Chernozhukov et al. \(2018\)](#). We distinguish Qwen3’s native thinking mode, toggled through the chat template, from chain-of-thought *prompting* ([Wei et al., 2022](#)); this paper tests only the former.

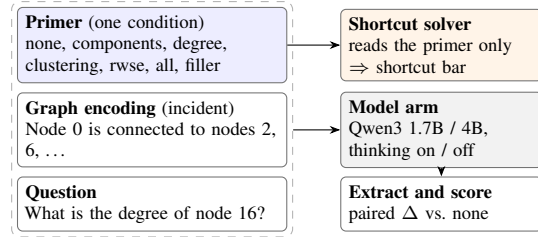


Figure 2: Experimental design. Each prompt joins a primer, the incident encoding and the question. The graph-blind shortcut solver reads only the primer and sets the bar a primer must exceed; *filler* controls length, *components* is a minimal-length content control, and rewiring (Section 3.3) changes the graph’s triangles while holding degrees and prompt length fixed.

## 3 Method

### 3.1 Primers

A primer is a preamble of factual sentences about a graph’s structure, prepended to the standard text encoding and the task question (Figure 2; a complete example appears in Appendix A.1). A single renderer produces all seven conditions, so conditions share sentence framing and number formatting and differ only in the facts stated. The exception is *filler*, which states no fact and therefore uses its own frame (“Node  $N$  is simply present within the graph  $G$ .”); it is a length control, not a content control. *none* is the graph encoding alone. *components* gives one sentence stating the number of connected components. *degree* states each node’s degree, *clustering* each node’s clustering coefficient, and *rwse* each node’s random-walk return probabilities after two and three steps ([Dwivedi et al., 2022](#)). *all* combines degree, clustering and *rwse*. Because the model reads tokens rather than characters, we measure length with the Qwen3 tokenizer: mean primer length at  $n=40$  is 8 tokens (*components*), 372 (*degree*), 470 (*filler*), 510 (*clustering*), 1,070 (*rwse*) and 1,612 (*all*), against a mean *none* prompt of 2,165 tokens. *filler* is therefore longer than *degree*, 8% shorter than *clustering* (a conservative control for it), and substantially shorter than *rwse* and *all*, for which it bounds rather than matches the length term.

### 3.2 Shortcut solvers

A primer’s measured benefit is evidence of graph reasoning only if it exceeds what a solver that never sees the graph can recover from the primer text. Our shortcut solver combines sixteen exact

rules, one heuristic, eight fitted rules (fit and scored on disjoint graph sets) and an exact enumeration bound for small graphs. Because it has no access to the graph object, its accuracy on a (task, condition) cell is a content-only lower bound, which we call the *shortcut bar*: it never overstates what the primer text alone permits, but it does not upper-bound model accuracy.

The published GraphQA generator draws graphs of 5–19 nodes, whereas our evaluation corpus has  $n=40$ , and a bar fit on small graphs does not transfer. Refitting the solver on  $n=40$  Erdős–Rényi graphs at the seven densities of our sweep (disjoint 300-graph fit and test sets per density) raises the `edge_existence/none` bar from 0.498 to 0.735, an effect of class imbalance rather than primer content, and reduces the `degree-over-none` gap on that task from +0.296 to +0.011. We classify a (task, condition) pair as offering a *substitute route* when its bar exceeds the `none` bar by more than 0.05; at  $n=40$  every gain is either at most 0.011 or at least 0.066, so any threshold between these values yields the same split. Under the refit bars (Table 1, bottom), only `node_degree` (under `rwse`, `degree`, `all`) and `edge_count` (under `clustering`, `degree`, `all`) offer a route. `node_count` and `cycle_check` are excluded from the split: at  $n=40$  their gold answers are constant (40, and “yes” at all seven densities), so a blind solver reaches 1.00 under every condition. The published-split bars (Table 1, top) are used for the held-out arms, which answer that corpus.

### 3.3 Rewiring: isolating structure from length

The `filler` condition bounds the length found only approximately (Section 3.1), and comparisons across graph generators cannot isolate a structural quantity: across five generator families at matched  $(n, m)$ , the majority-class baseline alone ranges from 0.15 to 1.00. We therefore use a within-instance design. A double-edge swap removes edges  $(a, b)$  and  $(c, d)$  and adds  $(a, d)$  and  $(c, b)$  (Fosdick et al., 2018); every endpoint keeps its degree, so the degree sequence,  $n$ ,  $m$ , the gold answer and the prompt length are unchanged, while the triangle count, which the `clustering` primer reports, changes. We generate low, base and high triangle-count variants of each graph ( $n=40$ , mean degree 12, 200 graphs per level; mean clustering coefficient 0.01, 0.31 and 0.73) and evaluate `node_degree` under `none`,

| Task (5–19 nodes)                                                                 | none | comp.       | clust.      | rwse        | degree      | filler      | all         |
|-----------------------------------------------------------------------------------|------|-------------|-------------|-------------|-------------|-------------|-------------|
| <code>node_count</code>                                                           | .064 | .076        | <b>1.00</b> | <b>1.00</b> | <b>1.00</b> | <b>1.00</b> | <b>1.00</b> |
| <code>edge_count</code>                                                           | .018 | .018        | .148        | .018        | <b>1.00</b> | .018        | <b>1.00</b> |
| <code>node_degree</code>                                                          | .082 | .082        | .082        | .616        | <b>1.00</b> | .082        | <b>1.00</b> |
| <code>connected_nodes</code>                                                      | .082 | .082        | .082        | .082        | .208        | .082        | .352        |
| <code>edge_existence</code>                                                       | .498 | .498        | .720        | .642        | .794        | .498        | .794        |
| <code>cycle_check</code>                                                          | .832 | <b>1.00</b> | .832        | .832        | .946        | .832        | .946        |
| <i>Task (<math>n=40</math>, mean over <math>p \in \{.10, \dots, .85\}</math>)</i> |      |             |             |             |             |             |             |
| <code>node_count</code>                                                           | 1.00 | 1.00        | 1.00        | 1.00        | 1.00        | 1.00        | 1.00        |
| <code>edge_count</code>                                                           | .030 | .030        | .104        | .030        | <b>1.00</b> | .030        | <b>1.00</b> |
| <code>node_degree</code>                                                          | .144 | .144        | .144        | .210        | <b>1.00</b> | .144        | <b>1.00</b> |
| <code>connected_nodes</code>                                                      | .000 | .000        | .000        | .001        | .001        | .000        | .008        |
| <code>edge_existence</code>                                                       | .735 | .735        | .727        | .554        | .746        | .735        | .741        |
| <code>cycle_check</code>                                                          | 1.00 | 1.00        | 1.00        | 1.00        | 1.00        | 1.00        | 1.00        |

Table 1: Shortcut bars: accuracy of a solver that reads only the rendered primer, never the graph. Top: 500 graphs from the published generator, used for the held-out arms. Bottom: refit at  $n=40$ , the size evaluated in the main sweep, averaged over seven densities. **Bold**: the primer content alone determines the answer. The unbolded 1.00 entries at  $n=40$  reflect a constant gold answer, not primer content.

`clustering` and `filler` with qwen3-1.7b, qwen3-1.7b-think and qwen35-2b (Section 4).

The design tests two predictions. If the `clustering` benefit derives from the stated values, it should vary with triangle count when everything else is fixed. More specifically, local clustering in an Erdős–Rényi graph is approximately the edge probability  $p$ , so the primer implicitly states the density, and  $\bar{C}(n-1)$  approximates the expected degree. A model that used the primer as a density prior would bias its degree estimates toward  $\bar{C}(n-1)$ , which equals 0.5, 12.0 and 28.7 across the three levels while the true mean degree stays at 12.

### 3.4 Answer extraction and measurement validity

Answer extraction errs asymmetrically across conditions, because thinking responses are longer and less regular than direct ones. One failure produced a spurious effect of headline magnitude: the boolean extractor took the last yes/no token, and on `cycle_check` qwen3-4b often answers correctly and then recites the definition of a cycle (“... a path with no repeated edges or nodes”). Ignoring “no repeated” moved 225 of qwen3-4b’s rows from incorrect to correct and none in the other direction, and removed an apparent condition-dependent collapse; all results use the corrected extractor. A manual check of 50 rows per arm and a corpus-wide comparison found that `<think>` blocks never reach the extractor unstripped. Scor-

ing truncated generations as incorrect instead of dropping them changes the significance verdict on three qwen3-1.7b cycle\_check cells (rwse, clustering, all; Appendix A); the main text drops truncated pairs.

### 3.5 Significance testing

Every comparison is paired: a condition against none (or another condition) on the same graph and question, with pairs dropped when either side is truncated. The primary test is the exact McNemar test (McNemar, 1947), Benjamini–Hochberg corrected (Benjamini and Hochberg, 1995) within each (arm, task) family of six conditions at  $q=0.05$ ; where we claim that a condition beats both controls, we additionally require significance under a global correction across every test run for that claim. For the relation between effect and baseline (Section 5.4), baseline and effect are estimated from disjoint random halves of each cell’s paired graphs, and the difference between route and no-route cells is tested directly as the interaction in  $\Delta \sim \text{baseline} \times \text{route}$ , with a bootstrap that resamples whole (arm, task, density) blocks, because conditions within a block share one none sample. For each null cell we estimate a minimum detectable effect (MDE): the smallest paired effect that the same test detects with probability 0.8 at  $\alpha=0.05$  in simulation (50 replicates, 200 permutations per replicate).

## 4 Experimental Setup

**Tasks.** Six GraphQA tasks (Fatemi et al., 2023): node count, edge count, node degree, connected nodes, edge existence and cycle check.

**Models.** An *arm* is a model checkpoint with a fixed decoding configuration. The main arms are Qwen3-1.7B and Qwen3-4B (Qwen Team, 2025), each with native thinking mode off (qwen3-1.7b, qwen3-4b) and on (-think), chosen because they offer both modes of the same weights at two sizes and moderate cost.

**Sweep design.** All four arms answer one shared prompt set: Erdős–Rényi graphs (Erdős and Rényi, 1959) with  $n=40$  at densities  $p \in \{.10, .20, .35, .50\}$ , 100 graphs per (density, task) cell, seven conditions and six tasks, for 16,800 prompts per arm. Decoding is greedy with a budget of 2,048 new tokens for plain arms and 8,192 for thinking arms; Section 5.6 reports a budget-matched comparison. A high-density extension

| Task            | 1.7B  | 1.7B-th.     | 4B   | 4B-th.       |
|-----------------|-------|--------------|------|--------------|
| edge_count      | 22.4% | <b>81.0%</b> | 0.6% | <b>62.2%</b> |
| cycle_check     | 6.1%  | 6.5%         | 1.0% | 1.9%         |
| node_degree     | 0.0%  | 1.1%         | 0.0% | 1.6%         |
| connected_nodes | 0.0%  | 0.8%         | 0.0% | 3.5%         |
| edge_existence  | 0.0%  | 0.2%         | 0.1% | 0.8%         |
| node_count      | 0.0%  | 2.5%         | 0.0% | 0.1%         |

Table 2: Truncation rate (share of main-sweep generations reaching the output budget: 2,048 tokens plain, 8,192 thinking), by task and arm, pooled over condition and density. **Bold:** above 50%.

covers node\_degree and edge\_existence at  $p \in \{.65, .75, .85\}$  (100 graphs per cell, all four arms). For the one content effect that survives the controls (Section 5.5), a dedicated node\_degree sweep with qwen3-1.7b uses 400 graphs per density at all seven densities, and a replication uses 400 newly seeded graphs per density at  $p \leq .50$ .

**Held-out arms.** Eleven further arms were not used to develop any claim reported here. Eight (Qwen3-8B, Qwen3-14B, Gemma-4-E4B-it and Gemma-4-12B-it (Google DeepMind, 2026), each with and without thinking) answer the published GraphQA split (30 graphs per cell, five tasks, six conditions); three (Qwen3-0.6B with and without thinking, and Qwen3.5-2B (Qwen Team, 2026), qwen3.5-2b) answer a 100-graph probe of none and degree on all six tasks. The published-split responses were collected before this project’s hypotheses were formed and were analysed earlier for other questions, so we describe these arms as held out rather than pre-registered.

**Metrics.** Exact match for every task. For connected\_nodes, set-F1 is 0.96–1.00 in every cell at this graph size and does not discriminate between conditions, so we report exact match. For the integer tasks, mean absolute error is reported in the Appendix.

## 5 Results

All four arms completed the main sweep (16,800 scored generations each). Unless stated otherwise,  $\Delta$  is a paired difference against none, in percentage points, over instances in which neither condition was truncated.

### 5.1 Truncation and constant answers

On edge\_count the thinking arms truncate on 62–81% of instances (Table 2), so surviving pairs



are a selected subset that we report but do not interpret; every other task truncates on fewer than 7%. `node_count` (gold = 40) and `cycle_check` (gold = “yes” at every density) have constant answers (Section 3.2); their apparent gains are reported in Table 3 but interpreted as artifacts (Section 5.3), not as graph reasoning.

## 5.2 The full density sweep

Table 3 (thinking arms in Table 4) pools over the four densities; paired bootstrap intervals for every cell are given in the Appendix. Several cells are significant in one arm without replicating in the other with the same sign and magnitude; Section 5.4 shows why this is expected, since the two model sizes are rarely at matched baseline accuracy on the same cell. Mean absolute error (Appendix Table 13) agrees in sign, except that `degree`’s +27.7 on `edge_count` for qwen3-4b leaves it unchanged (39.2 vs. 40.4 edges).

## 5.3 Length is an active treatment

`filler` is not inert. It significantly reduces accuracy for qwen3-1.7b on `edge_existence` (−15.8) and `connected_nodes` (−6.2), and for qwen3-4b on both (−7.0, −12.5). Over the full seven-density range (qwen3-1.7b-think, `node_degree`) it costs 6.3 points pooled ( $p < 10^{-4}$ ): negligibly at  $p \leq .35$ , then 9.5, 9.3, 14.1 and 11.7 points at the four densest levels ( $r = +0.953$  with baseline across levels,  $p = 0.00088$ ). The cost of added length thus grows with task difficulty, in the same direction as the benefit of an answer-stating primer (Section 5.4).

The largest `edge_existence` effects in Table 3 do not reflect more accurate reading of the graph. Appendix Table 7 decomposes accuracy into hit rate and false-alarm rate: the hit rate is 99–100% under every condition for both arms, so accuracy varies only through false alarms, which `filler` increases (40.3→61.8% and 12.6→22.5%) and `clustering` decreases. A signal-detection analysis (Macmillan and Creelman, 2005), with the log-linear correction for extreme proportions (Hautus, 1995), cannot separate a shift in response criterion from a change in sensitivity here: with the hit rate at ceiling, a change in the false-alarm rate alone moves  $d'$  and  $c$  together and cannot be attributed to either (Appendix Table 10). We therefore report only that accuracy on this task moves through the false-alarm rate, and do not interpret these effects as changes in graph reading. Under

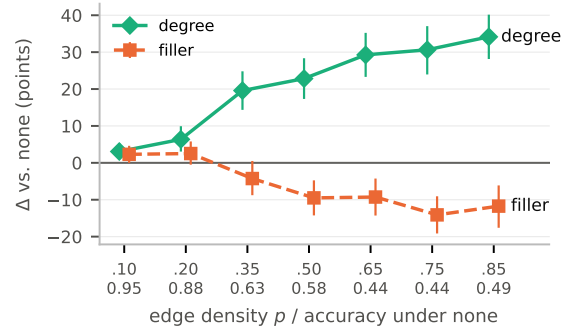


Figure 3: Density continuum, qwen3-1.7b-think, `node_degree`: paired effect against none (400 graphs per density, 95% bootstrap intervals). Lower tick row: accuracy under none.

`none`, qwen3-1.7b’s criterion becomes steadily more liberal with density ( $c$  from −0.27 at  $p = .10$  to −1.74 at  $p = .50$ ).

`node_count`’s largest apparent gains are also an artifact. Gold is the constant 40; the dominant `none` error is “39”, an off-by-one induced by the encoder’s “nodes 0, ..., 39” preamble. qwen3-1.7b answers “39” on 68.5% of instances under `none`, 61.8% under `degree`, 28.8% under `filler` and 1.3% under `clustering`. Almost any interposed text reduces this error, and none of these changes is a structural effect on a task with a single possible answer.

## 5.4 Answer-stating primers: route substitution

A *cell* is an (arm, task, density, condition) combination with at least ten usable pairs. Appendix Table 5 splits the 354 main-sweep cells on non-constant tasks by whether the primer offers a substitute route. Route cells gain at low baseline and lose at high baseline ( $r = -0.417$ , 81 cells); no-route cells do not ( $r = -0.050$ , 273 cells). Two problems prevent reading this contrast directly: baseline and effect share sampling noise, and a difference in significance is not a significant difference (Gelman and Stern, 2006). We therefore estimate baseline and effect from disjoint halves of each cell’s graphs and test the route  $\times$  baseline interaction, adding the high-density extension (117 route and 381 no-route cells; Figure 1). The interaction is −11.0 points per unit of baseline (95% block-bootstrap CI [−17.1, −5.7], 83 blocks). Within groups, the cross-fitted correlation is  $r = -0.34$  for route cells (116 cells, one fold direction,  $p = 1.6 \times 10^{-4}$ ) and

| Arm        | Task            | none | comp.        | clust.       | rwse         | degree       | filler       | all          |
|------------|-----------------|------|--------------|--------------|--------------|--------------|--------------|--------------|
| qwen3-1.7b | connected_nodes | 72.5 | +0.8         | +1.2         | +1.0         | +3.5         | <b>-6.2</b>  | +2.0         |
|            | cycle_check     | 99.7 | -0.6         | +0.3         | <b>-7.5</b>  | <b>-12.6</b> | +0.3         | -1.1         |
|            | edge_count      | 1.8  | -1.4         | -0.4         | -0.8         | +4.0         | -0.4         | +2.4         |
|            | edge_existence  | 70.5 | <b>-8.2</b>  | +3.8         | +5.3         | +4.0         | <b>-15.8</b> | <b>+11.5</b> |
|            | node_count      | 31.5 | <b>+21.2</b> | <b>+67.2</b> | +2.0         | <b>+6.8</b>  | <b>+39.8</b> | <b>+47.0</b> |
|            | node_degree     | 60.5 | +1.8         | +3.0         | +4.0         | <b>+7.5</b>  | -0.5         | <b>+10.2</b> |
| qwen3-4b   | connected_nodes | 86.0 | <b>+9.2</b>  | +1.0         | +0.0         | -3.0         | <b>-12.5</b> | -2.8         |
|            | cycle_check     | 99.7 | -1.3         | +0.3         | +0.3         | -1.5         | -0.3         | -0.5         |
|            | edge_count      | 2.0  | -1.3         | -1.3         | -1.5         | <b>+27.7</b> | +1.2         | <b>+3.8</b>  |
|            | edge_existence  | 90.5 | -2.8         | <b>+4.0</b>  | <b>-10.0</b> | <b>-4.3</b>  | <b>-7.0</b>  | <b>-9.8</b>  |
|            | node_count      | 90.2 | <b>+9.0</b>  | <b>+8.8</b>  | <b>+9.2</b>  | <b>+8.0</b>  | <b>+9.2</b>  | <b>+9.8</b>  |
|            | node_degree     | 99.2 | +0.2         | -0.5         | <b>-2.2</b>  | <b>-6.5</b>  | -0.5         | <b>-8.0</b>  |

Table 3: Main sweep, plain arms (thinking arms in Table 4),  $n=40$ , pooled over densities  $p \in \{0.10, 0.20, 0.35, 0.50\}$ , 100 graphs per (task, primer, density) cell. The none column is absolute accuracy (%); all other columns are paired differences against none in percentage points. Truncated generations excluded pairwise,  $n \in [211, 400]$  paired instances/cell. **Bold**: significant, exact McNemar, Benjamini–Hochberg within each (arm, task) family,  $q = 0.05$ .

| Arm              | Task            | none  | comp. | clust. | rwse        | degree       | filler | all          |
|------------------|-----------------|-------|-------|--------|-------------|--------------|--------|--------------|
| qwen3-1.7b-think | connected_nodes | 79.7  | +2.8  | +1.5   | -0.3        | <b>+7.4</b>  | +2.0   | <b>+6.1</b>  |
|                  | cycle_check     | 100.0 | +0.0  | +0.0   | +0.0        | -0.7         | +0.0   | +0.0         |
|                  | edge_count      | 86.3  | +7.4  | -8.7   | -4.0        | <b>-61.5</b> | +3.4   | -8.3         |
|                  | edge_existence  | 99.2  | -0.8  | -1.0   | -1.3        | -2.3         | -0.8   | -2.0         |
|                  | node_count      | 98.6  | +0.9  | +1.4   | +1.4        | +0.8         | +1.4   | +1.4         |
|                  | node_degree     | 76.8  | -1.0  | +1.0   | +2.5        | <b>+11.7</b> | -3.5   | <b>+11.6</b> |
| qwen3-4b-think   | connected_nodes | 98.7  | +0.5  | +0.0   | +0.0        | +0.0         | +0.8   | -0.5         |
|                  | cycle_check     | 100.0 | +0.0  | +0.0   | +0.0        | -0.8         | +0.0   | +0.0         |
|                  | edge_count      | 97.6  | +1.7  | +1.7   | +0.0        | <b>-23.5</b> | -2.0   | <b>-41.5</b> |
|                  | edge_existence  | 100.0 | +0.0  | +0.0   | +0.0        | +0.0         | +0.0   | +0.0         |
|                  | node_count      | 99.5  | +0.0  | +0.5   | +0.5        | +0.5         | +0.5   | +0.5         |
|                  | node_degree     | 99.7  | -0.3  | -0.3   | <b>-2.1</b> | +0.0         | +0.0   | -0.8         |

Table 4: Main sweep, thinking arms; conventions as in Table 3. Both arms are at or near ceiling on most tasks, leaving no headroom for a primer effect. edge\_count truncates on 62–81% of rows under thinking (Table 2) and is reported but not interpreted.  $n \in [13, 400]$  paired instances/cell. **Bold**: significant, exact McNemar, Benjamini–Hochberg within each (arm, task) family,  $q = 0.05$ .

$r=+0.03$  for no-route cells; the other fold direction gives  $-0.28$  and  $+0.02$ .

Accuracy is bounded, so a cell near 1.0 can only lose and a cell near 0.0 can only gain. On the logit scale, which removes this compression, the interaction persists ( $-1.46$  logits, CI  $[-2.11, -0.79]$ ). Restricted to cells with baseline between 0.10 and 0.90, however, neither group shows a relation (route  $r=-0.01$ , 78 cells; no-route  $r=-0.07$ , 187 cells; interaction CI  $[-29, +41]$ ). The evidence therefore supports a contrast between floor and ceiling (a stated answer helps a model that cannot compute it and harms one that already can) rather than a graded dependence that can be detected at intermediate baselines, where our data are sparsest (Appendix Table 8).

The density continuum (Figure 3) manipulates difficulty instead of observing it. Holding the arm (qwen3-1.7b-think) and task

(node\_degree) fixed and raising density from  $p=.10$  to  $.85$  lowers none accuracy from 0.95 to 0.44, while degree’s benefit rises from  $+3.1$  to  $+34.2$  points ( $r=-0.975$ ,  $p=1.9 \times 10^{-4}$ ) and filler’s cost grows to 14.1 points.

**Held-out arms and a second family.** On the eleven held-out arms the relation reappears where a route exists:  $r=-0.674$  over 105 route cells ( $p=3.5 \times 10^{-15}$ ) against  $r=-0.186$  over 93 no-route cells ( $p=0.075$ ), with an interaction of  $-24.7$  (block-bootstrap CI  $[-109, -3.8]$ ,  $p=0.027$ , 43 (arm, task) blocks), weaker than on the main sweep but excluding zero. In an ordinary least-squares regression of effect on baseline, log parameter count and thinking mode (classical standard errors, so descriptive only), the baseline coefficient is  $-30.2$  and the parameter coefficient  $+0.40$  ( $p=0.54$ ): once baseline is accounted for, scale adds nothing measurable. Replacing each cell’s own baseline

with the other arms’ accuracy on the same cell, a negative control that shares no observation with the effect, removes the relation ( $r=+0.113$ ,  $p=0.135$ ), so the effect tracks a model’s own competence rather than item difficulty. Most held-out arms are near ceiling under *none* (mean 0.924), and a confirmatory 500-graph run with a pre-specified condition (qwen3-8b, *degree*) gives +6.67 points ( $p=1\times 10^{-4}$ ).

**Renaming nodes.** Replacing integer node identifiers with Game-of-Thrones character names throughout the prompt leaves the route effect largely intact: on the published split, *degree*’s effect correlates at  $r=0.933$  between the integer- and name-based sweeps (39 cells), whereas non-route conditions correlate weakly or not at all (*clustering*  $r=0.079$ , *components*  $r=0.345$ , *rwse*  $r=-0.381$ ). Renaming without a primer changes accuracy by  $-0.19$  points pooled across eight arms (*none* significant after correction). Because renaming changes identifiers but not the stated degree values, this shows that the effect does not depend on integer labels; it does not distinguish copying the stated value from computing the answer, which requires a primer that states incorrect values (Section 7).

**Error analysis.** When a primer states the answer and the model still errs, the errors indicate positional copying. Of qwen3-4b’s 29 errors under *degree* on *node\_degree*, 12 (41.4%) equal the stated degree of the node listed immediately before or after the queried one, against 17.3% expected by chance (binomial  $p=.002$ ); under *all*, which states the same degree amid three facts per node, the rate is 2 of 35 (5.7%).

## 5.5 Non-answer content: clustering on node\_degree

*clustering* on *node\_degree* is the only cell in which a primer that does not state the answer was followed up. It was selected after the main sweep and examined in a dedicated sweep and an independent-seed replication (Figure 4; Appendix Table 6). Against *none*, the effect is small when pooled over seven densities (+1.9 points,  $p=0.022$ ) and significant individually only at  $p=.35$  (+7.5); it is +0.3 and +3.3 at the two lowest densities, +4.3 at  $p=.50$ , and within two points of zero at the three highest. The independent-seed replication at  $p\leq .50$  confirms a modest gain (+4.2,  $p=0.00055$ ). Against the length-matched *filler*, the pooled

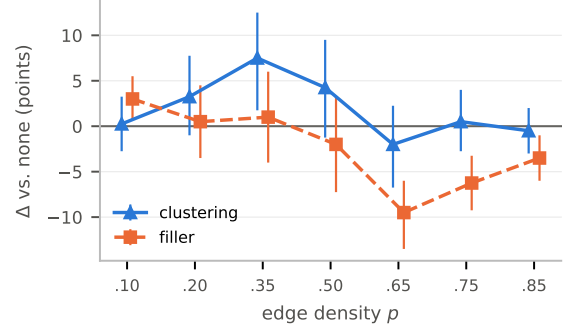


Figure 4: clustering and filler against *none* on *node\_degree*, qwen3-1.7b, by density (400 graphs per density, 95% bootstrap intervals). Pooled comparisons: Appendix Table 6.

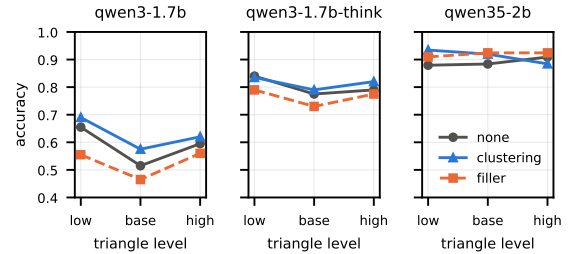


Figure 5: Rewiring: *node\_degree* accuracy by triangle level (200 graphs per level; degree sequence and prompt length fixed; mean clustering 0.01, 0.31, 0.73).

advantage is larger ( $+4.3$ ,  $p=2.4\times 10^{-8}$ ), but much of it reflects *filler*’s own cost ( $-2.4$  against *none*,  $p=0.0013$ ). At  $p\geq .65$ , where *clustering* and *none* do not differ, *filler* costs 3.5–9.5 points, which accounts for the entire difference there. Only at  $p=.35$  and  $.50$  does *clustering* exceed both controls (+6.5 and +6.3 against *filler*,  $p=0.027$  and  $0.022$ ). Length does not explain this residual effect, since *filler* is 40 tokens shorter than *clustering*. For qwen3-4b, which is at ceiling on this task in the main sweep, the high-density extension gives +11.3 points against *none* ( $p<10^{-3}$ ) and +9.3 against *filler* ( $p=6\times 10^{-4}$ ), both significant under the global correction. The effect is absent in qwen3-1.7b-think (Table 4), and qwen3-4b-think is at ceiling on this task throughout.

The rewiring experiment constrains the mechanism but does not identify it (Figure 5). Mean clustering rises from 0.01 to 0.31 to 0.73 across levels by construction, but the benefit does not follow: for qwen3-1.7b, clustering minus *none* is +3.5, +6.0 and +2.5 points at the three levels,

broadly parallel to `none`’s own level-to-level variation, and for `qwen35-2b` `filler` matches or exceeds `clustering` at every level. The density-prior prediction (Section 3.3) fails as well: it implies that errors under `clustering` should shift upward by up to 17 at the `high` level, but the paired shift in signed error relative to `none` is within 0.3 of zero at every level for all three arms (Appendix Table 12). Neither mechanism we can test accounts for the effect.

## 5.6 Thinking mode

Thinking brings most cells to ceiling, and where it does not, truncation censors the task (Table 2). Generation length also depends on condition: for `qwen3-1.7b-think`, `degree` raises the non-termination rate on `node_degree` by 3.4 points ( $p < 0.001$ ), producing 130 non-terminating generations outside `edge_count` that do not occur under `none`. `qwen3-4b-think` answers correctly on all 29 instances on which `degree` causes `qwen3-4b` to fail.

Because the two modes had different budgets, we rescored the thinking arms as if capped at the plain arms’ 2,048 tokens; under greedy decoding this is exact, since a trace that finished within 2,048 tokens is unaffected by the lower cap. At equal budget, thinking still outperforms the plain mode on `edge_existence` (97.8 vs. 70.5% for 1.7B; 99.8 vs. 90.5% for 4B) and `node_count`, but underperforms on `cycle_check` (38.0 vs. 90.7%; 62.0 vs. 99.0%) and on `node_degree` for 4B (91.2 vs. 99.3%), and no `edge_count` trace finishes within the budget (Appendix Table 11). Part of the advantage of thinking is therefore an advantage of budget.

## 5.7 Statistical power

For every null cell in the main sweep we estimate an MDE (Section 3.5). Where it is finite, the median MDE is 5.6 (`qwen3-1.7b`) and 10.0 points (`qwen3-4b`) for gains and 2–3 points for losses. In 10 of 24 null cells for `qwen3-1.7b`, 10 of 20 for `qwen3-4b`, 23 of 33 for `qwen3-1.7b-think` and 33 of 33 for `qwen3-4b-think`, an effect in at least one direction is undetectable at any magnitude because `none` accuracy is at or near 0 or 100%. These nulls reflect ceiling or floor, not evidence of no effect.

## 6 Discussion

Three regularities account for most of the measured effects. For RQ1, a primer that states the requested quantity acts as a second route to the answer, helping a model at floor and harming one at ceiling; the contrast survives cross-fitting, a logit transformation, held-out arms, a negative-control difficulty estimate and node renaming, but it is not graded at intermediate baselines, and renaming cannot establish whether the model copies or computes. For RQ2, added text is an active treatment whose cost exceeds most content effects and grows with difficulty, so a primer compared only against `none` confounds content with length. For RQ3, one effect of a primer that does not state the answer survives, `clustering` on `node_degree`, worth four to seven points at intermediate densities for `qwen3-1.7b` and present in `qwen3-4b` below ceiling, but neither mechanism we could test explains it. All claims are behavioural: we measured no internal states of the models.

## 7 Future Work

Each of the following needs new generations. A *shuffled-value* primer (same template and values, permuted across nodes) is a stricter placebo than *filler*; a *corrupted* primer stating wrong degrees separates copying from computation; and a *single-fact* primer separates interference from route substitution. Varying primer position, encoding and generator family (e.g. Barabási–Albert, in which clustering varies across nodes), matching fillers in tokens to each condition, running the plain arms at 8,192 tokens, and extending the `clustering` sweep across Qwen3 sizes would address the remaining limitations.

## 8 Conclusion

Structural primers change LLM accuracy on GraphQA mainly by stating the answer, which helps only a model that cannot otherwise produce it, and by lengthening the prompt, which costs accuracy on harder instances; the one surviving effect of non-answer content is small and unexplained. Evaluations of prompt-level graph augmentations should report baseline accuracy, a length-matched control and a graph-blind solver.

## Limitations

The main sweep covers two model sizes from one family; the held-out arms extend the range but



on a smaller corpus with fewer conditions and were not pre-registered. The route classification depends on a threshold and on the  $n=40$  shortcut refit, and the route relation is not detectable within the 0.10–0.90 baseline window, where data are sparse. All graphs are Erdős–Rényi, in which local clustering is nearly uniform, so the `clustering` primer carries little node-specific information; the rewiring varies triangle count within this family but does not test other families. The length control matches `clustering` to within 8% in tokens but is much shorter than `rwse` and `all`. Every prompt uses one encoding (`incident`), one template and one primer position, with greedy decoding. The `clustering` cell was selected after the main sweep, and several results rest on 400 or fewer instances per cell; an MDE is reported for every null result, but it does not replace a larger sample.

## Ethics Statement

This work evaluates open-weight language models on synthetically generated Erdős–Rényi graphs. It involves no human subjects, no personal or identifying data and no annotation labour. We see no direct risk of harm from the findings; the principal cost is the GPU compute for approximately 200,000 generations.

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## A Supplementary material

### A.1 Example prompt

A clustering-condition node\_degree prompt at  $p=0.10$  (elisions marked ...). Other conditions replace the first paragraph; none omits it.

Node 0 has clustering coefficient 0.20. Node 1 has clustering coefficient 0.00. ... Node 39 has clustering coefficient 0.00.

G describes a graph among nodes 0, 1, 2, ..., 38, and 39.

In this graph:

Node 0 is connected to nodes 2, 6, 8, 12, 35.  
Node 1 is connected to nodes 5, 20.

...

Node 39 is connected to nodes 15, 16, 18, 20.

Q: What is the degree of node 16?

A:

### A.2 Robustness analyses

| Accuracy under<br>none       | substitute route |          | no route |          |
|------------------------------|------------------|----------|----------|----------|
|                              | cells            | $\Delta$ | cells    | $\Delta$ |
| 0.00–0.25                    | 24               | +6.3     | 24       | −0.7     |
| 0.25–0.50                    | 6                | +8.5     | 6        | +2.0     |
| 0.50–0.75                    | 6                | +15.5    | 24       | +0.4     |
| 0.75–0.95                    | 11               | +4.2     | 33       | −3.9     |
| 0.95–1.00                    | 34               | −9.0     | 186      | −0.4     |
| $r(\Delta, \text{baseline})$ | −0.417           |          | −0.050   |          |

Table 5: Mean primer effect (percentage points), binned by none accuracy on the same (arm, task, density, condition) cell of the main sweep, using the  $n=40$  route split (Table 1, bottom). Mean baseline is 0.61 for route cells and 0.85 for no-route cells, so the groups are not baseline-matched; the cross-fitted interaction test in the text addresses this.

| Comparison                 | $\Delta$ | $n$  | density | $p$                  |
|----------------------------|----------|------|---------|----------------------|
| clust. vs. none (main)     | +3.0     | 400  | .10–.50 | 0.26                 |
| clust. vs. none            | +1.9     | 2800 | .10–.85 | 0.022                |
| filler vs. none            | −2.4     | 2800 | .10–.85 | 0.0013               |
| clust. vs. filler          | +4.3     | 2800 | .10–.85 | $2.4 \times 10^{-8}$ |
| clust. vs. none            | +7.5     | 400  | .35     | 0.0089               |
| clust. vs. none (new seed) | +4.2     | 1600 | .10–.50 | 0.00055              |

Table 6: qwen3-1.7b, node\_degree. All rows except the first come from the dedicated 400-graph-per-density sweep; the last row uses newly seeded graphs. Per-density values are in Appendix Table 9.

| Arm        | Cond.      | hit rate | false alarm |
|------------|------------|----------|-------------|
| qwen3-1.7b | none       | 100.0%   | 40.3%       |
|            | filler     | 100.0%   | 61.8%       |
|            | clustering | 100.0%   | 35.2%       |
| qwen3-4b   | none       | 99.1%    | 12.6%       |
|            | filler     | 100.0%   | 22.5%       |
|            | clustering | 100.0%   | 7.5%        |

Table 7: edge\_existence,  $p=0.50$ : hit rate (correct on gold “yes”) and false-alarm rate (“yes” on gold “no”). Accuracy changes almost entirely through the false-alarm rate.

| Estimate                      | $r_{\text{route}}$ | $k$ | $r_{\text{none}}$ | $k$ | interaction [95% CI] |
|-------------------------------|--------------------|-----|-------------------|-----|----------------------|
| naive, pp                     | -0.35              | 117 | -0.01             | 381 | -12.5 [-19.5, -6.3]  |
| cross-fit (one direction), pp | -0.34              | 116 | +0.03             | 381 | -12.1 [-17.9, -7.0]  |
| cross-fit (both), pp          | -0.31              | 232 | +0.02             | 762 | -11.0 [-17.1, -5.7]  |
| cross-fit (both), logit       | -0.40              | 232 | +0.04             | 762 | -1.35 [-1.93, -0.77] |
| baseline .10-.90, pp          | -0.01              | 78  | -0.07             | 187 | +3.0 [-29.1, +41.1]  |
| baseline .10-.90, logit       | +0.02              | 78  | -0.05             | 187 | +0.22 [-1.36, +1.94] |
| held-out, pp                  | -0.67              | 105 | -0.19             | 93  | -24.7 [-109, -3.8]   |

Table 8: Route  $\times$  baseline relation, main sweep plus high-density extension (held-out: published split and probe). “Both” counts each cell once per fold direction;  $k$  for the window rows counts points. Interaction CIs resample (arm, task, density) blocks, or (arm, task) for held-out arms.

| $p$ | clust.-none | filler-none | clust.-filler |
|-----|-------------|-------------|---------------|
| .10 | +0.3        | +3.0*       | -2.8*         |
| .20 | +3.3        | +0.5        | +2.8          |
| .35 | +7.5*       | +1.0        | +6.5*         |
| .50 | +4.3        | -2.0        | +6.3*         |
| .65 | -2.0        | -9.5*       | +7.5*         |
| .75 | +0.5        | -6.3*       | +6.8*         |
| .85 | -0.5        | -3.5*       | +3.0*         |

Table 9: gwen3-1.7b, node\_degree, dedicated sweep, 400 graphs per density. \*: McNemar  $p < 0.05$  (uncorrected).

| Cond.      | gwen3-1.7b           |            |                      | gwen3-4b    |            |  |
|------------|----------------------|------------|----------------------|-------------|------------|--|
|            | $\Delta d'$          | $\Delta c$ |                      | $\Delta d'$ | $\Delta c$ |  |
| components | -0.36 [-0.56, -0.21] | -0.18      | +0.07 [-0.24, +0.33] | -0.10       |            |  |
| clustering | +0.21 [+0.03, +0.41] | +0.11      | +0.43 [+0.12, +0.75] | +0.08       |            |  |
| rwse       | +0.07 [-0.19, +0.43] | +0.28      | -0.38 [-0.68, -0.15] | -0.32       |            |  |
| degree     | +0.15 [-0.05, +0.41] | +0.19      | -0.04 [-0.33, +0.21] | -0.15       |            |  |
| filler     | -0.93 [-1.16, -0.71] | -0.46      | -0.25 [-0.54, +0.00] | -0.25       |            |  |
| all        | +0.55 [+0.35, +0.77] | +0.27      | -0.36 [-0.71, -0.11] | -0.31       |            |  |

Table 10: edge\_existence, main sweep: change in sensitivity  $d'$  and criterion  $c$  against none, averaged over the four densities (log-linear correction; 95% bootstrap CI for  $\Delta d'$ ). With the hit rate at ceiling,  $d'$  and  $c$  are not separately identifiable.

| Task            | 1.7B, acc. under none (%) |       |            | 4B, acc. under none (%) |       |            |
|-----------------|---------------------------|-------|------------|-------------------------|-------|------------|
|                 | plain                     | think | think@2048 | plain                   | think | think@2048 |
| connected_nodes | 72.5                      | 78.7  | 67.2       | 86.0                    | 95.5  | 89.0       |
| cycle_check     | 90.7                      | 97.0  | 38.0       | 99.0                    | 99.3  | 62.0       |
| edge_count      | 1.3                       | 15.8  | 0.0        | 2.0                     | 20.0  | 0.0        |
| edge_existence  | 70.5                      | 99.0  | 97.8       | 90.5                    | 99.8  | 99.8       |
| node_count      | 31.5                      | 91.0  | 86.0       | 90.2                    | 99.5  | 94.8       |
| node_degree     | 60.5                      | 76.2  | 67.7       | 99.3                    | 97.0  | 91.2       |

Table 11: Budget-matched comparison. Plain arms use 2,048 tokens; thinking arms 8,192 (“think”) or rescored at 2,048 (“think@2048”). Truncated generations are scored incorrect here, so values differ from the dropped-pair convention of Tables 3 and 4.

| Arm              | low   | base  | high  |
|------------------|-------|-------|-------|
| predicted shift  | -11.5 | 0.0   | +16.7 |
| gwen3-1.7b       | +0.03 | +0.28 | -0.02 |
| gwen3-1.7b-think | +0.01 | +0.04 | +0.01 |
| gwen35-2b        | -0.29 | -0.26 | -0.16 |

Table 12: Density-prior test on the rewiring corpus: mean paired shift in signed node\_degree error, clustering minus none (errors clipped at  $\pm 20$ , 200 graphs per level). “Predicted” is the shift if answers moved fully to  $\bar{C}(n-1)$ .

| Arm  | Task        | none  | comp. | clust. | rwse | degree | filler | all  |
|------|-------------|-------|-------|--------|------|--------|--------|------|
| 1.7B | node_count  | 0.69  | 0.47  | 0.01   | 0.67 | 0.62   | 0.29   | 0.22 |
|      | edge_count  | 133.4 | 98.1  | 105.7  | 79.0 | 47.7   | 181.2  | 94.2 |
|      | node_degree | 0.94  | 0.91  | 0.78   | 0.91 | 0.67   | 1.27   | 0.91 |
| 4B   | node_count  | 0.10  | 0.01  | 0.01   | 0.01 | 0.02   | 0.01   | 0.00 |
|      | edge_count  | 39.2  | 40.1  | 43.4   | 56.0 | 40.4   | 29.3   | 28.1 |
|      | node_degree | 0.02  | 0.01  | 0.02   | 0.05 | 0.14   | 0.02   | 0.49 |

Table 13: Mean absolute error on the integer tasks, plain arms, main sweep (non-truncated rows).

### A.3 Per-cell results

Paired effect sizes for every (arm, task, primer) cell:  $\Delta$  in percentage points against none, a 95% paired-bootstrap interval, the permutation  $p$ -value, and  $\Delta$  under the alternative convention that scores truncated generations incorrect rather than dropping the pair (Section 3.4).  $n$  is the number of pairs in which neither arm truncated.

| Task            | Primer     | $n$ | $\Delta$ | 95% CI         | perm $p$ | zeroed |
|-----------------|------------|-----|----------|----------------|----------|--------|
| connected_nodes | components | 400 | +0.8     | [-2.8, +4.5]   | 0.7861   | +0.8   |
|                 | clustering | 400 | +1.2     | [-3.0, +5.8]   | 0.6458   | +1.2   |
|                 | rwse       | 400 | +1.0     | [-3.5, +5.2]   | 0.7304   | +1.0   |
|                 | degree     | 400 | +3.5     | [-0.5, +7.5]   | 0.1239   | +3.5   |
|                 | filler     | 400 | -6.2     | [-10.2, -2.2]  | 0.0040   | -6.2   |
|                 | all        | 399 | +2.0     | [-2.8, +6.8]   | 0.4644   | +2.0   |
| cycle_check     | components | 343 | -0.6     | [-1.5, +0.0]   | 0.5033   | +2.0   |
|                 | clustering | 362 | +0.3     | [+0.0, +0.8]   | 1.0000   | +8.5   |
|                 | rwse       | 358 | -7.5     | [-10.3, -4.7]  | 0.0001   | -0.2   |
|                 | degree     | 301 | -12.6    | [-16.6, -9.0]  | 0.0001   | -17.5  |
|                 | filler     | 341 | +0.3     | [+0.0, +0.9]   | 1.0000   | +3.0   |
|                 | all        | 362 | -1.1     | [-2.5, +0.0]   | 0.2210   | +7.0   |
| edge_count      | components | 211 | -1.4     | [-3.8, +0.5]   | 0.3783   | -1.0   |
|                 | clustering | 238 | -0.4     | [-2.5, +1.7]   | 1.0000   | -0.5   |
|                 | rwse       | 239 | -0.8     | [-2.9, +1.3]   | 0.6837   | -0.8   |
|                 | degree     | 253 | +4.0     | [+0.8, +7.1]   | 0.0179   | +3.0   |
|                 | filler     | 224 | -0.4     | [-2.2, +1.3]   | 1.0000   | -0.5   |
|                 | all        | 211 | +2.4     | [-0.5, +5.7]   | 0.2269   | +0.8   |
| edge_existence  | components | 400 | -8.2     | [-11.8, -4.8]  | 0.0001   | -8.2   |
|                 | clustering | 400 | +3.8     | [-0.5, +8.0]   | 0.1199   | +3.8   |
|                 | rwse       | 399 | +5.3     | [+0.8, +9.8]   | 0.0321   | +5.0   |
|                 | degree     | 400 | +4.0     | [-0.5, +8.2]   | 0.0957   | +4.0   |
|                 | filler     | 400 | -15.8    | [-20.0, -11.8] | 0.0001   | -15.8  |
|                 | all        | 400 | +11.5    | [+6.8, +16.2]  | 0.0001   | +11.5  |
| node_count      | components | 400 | +21.2    | [+15.8, +26.8] | 0.0001   | +21.2  |
|                 | clustering | 400 | +67.2    | [+62.7, +71.8] | 0.0001   | +67.2  |
|                 | rwse       | 400 | +2.0     | [-4.2, +8.5]   | 0.5871   | +2.0   |
|                 | degree     | 400 | +6.8     | [+3.0, +10.5]  | 0.0008   | +6.8   |
|                 | filler     | 400 | +39.8    | [+35.0, +44.8] | 0.0001   | +39.8  |
|                 | all        | 400 | +47.0    | [+42.0, +52.0] | 0.0001   | +47.0  |
| node_degree     | components | 400 | +1.8     | [-2.0, +5.8]   | 0.4599   | +1.8   |
|                 | clustering | 400 | +3.0     | [-1.8, +7.8]   | 0.2577   | +3.0   |
|                 | rwse       | 399 | +4.0     | [-0.5, +8.5]   | 0.1104   | +4.0   |
|                 | degree     | 400 | +7.5     | [+3.0, +12.0]  | 0.0018   | +7.5   |
|                 | filler     | 400 | -0.5     | [-4.5, +3.5]   | 0.9028   | -0.5   |
|                 | all        | 400 | +10.2    | [+5.2, +15.2]  | 0.0003   | +10.2  |

Table 14: qwen3-1.7b: all cells.

| Task            | Primer     | $n$ | $\Delta$ | 95% CI         | perm $p$ | zeroed |
|-----------------|------------|-----|----------|----------------|----------|--------|
| connected_nodes | components | 392 | +2.8     | [-0.5, +6.1]   | 0.1268   | +2.5   |
|                 | clustering | 394 | +1.5     | [-2.5, +5.6]   | 0.5384   | +2.0   |
|                 | rwse       | 393 | -0.3     | [-4.1, +3.6]   | 1.0000   | +0.2   |
|                 | degree     | 390 | +7.4     | [+3.6, +11.3]  | 0.0001   | +6.2   |
|                 | filler     | 394 | +2.0     | [-1.5, +5.6]   | 0.3275   | +2.5   |
|                 | all        | 392 | +6.1     | [+2.6, +9.9]   | 0.0027   | +6.2   |
| cycle_check     | components | 346 | +0.0     | [+0.0, +0.0]   | 1.0000   | -7.5   |
|                 | clustering | 376 | +0.0     | [+0.0, +0.0]   | 1.0000   | -0.5   |
|                 | rwse       | 386 | +0.0     | [+0.0, +0.0]   | 1.0000   | +2.2   |
|                 | degree     | 295 | -0.7     | [-1.7, +0.0]   | 0.5036   | -21.8  |
|                 | filler     | 388 | +0.0     | [+0.0, +0.0]   | 1.0000   | +3.0   |
|                 | all        | 373 | +0.0     | [+0.0, +0.0]   | 1.0000   | -0.8   |
| edge_count      | components | 27  | +7.4     | [+0.0, +18.5]  | 0.5037   | -0.8   |
|                 | clustering | 23  | -8.7     | [-26.1, +8.7]  | 0.6251   | -1.2   |
|                 | rwse       | 25  | -4.0     | [-12.0, +0.0]  | 1.0000   | -3.2   |
|                 | degree     | 13  | -61.5    | [-84.6, -38.5] | 0.0102   | -3.2   |
|                 | filler     | 29  | +3.4     | [-6.9, +13.8]  | 1.0000   | -1.5   |
|                 | all        | 24  | -8.3     | [-29.2, +12.5] | 0.6794   | +5.0   |
| edge_existence  | components | 398 | -0.8     | [-2.3, +0.8]   | 0.5090   | -0.8   |
|                 | clustering | 398 | -1.0     | [-2.5, +0.3]   | 0.2911   | -1.0   |
|                 | rwse       | 398 | -1.3     | [-3.0, +0.3]   | 0.2274   | -1.2   |
|                 | degree     | 397 | -2.3     | [-4.3, -0.5]   | 0.0249   | -2.5   |
|                 | filler     | 399 | -0.8     | [-2.3, +0.8]   | 0.5051   | -0.5   |
|                 | all        | 399 | -2.0     | [-3.8, -0.5]   | 0.0374   | -1.8   |
| node_count      | components | 342 | +0.9     | [-0.3, +2.3]   | 0.3758   | +1.5   |
|                 | clustering | 369 | +1.4     | [+0.3, +2.7]   | 0.0597   | +9.0   |
|                 | rwse       | 369 | +1.4     | [+0.3, +2.7]   | 0.0597   | +9.0   |
|                 | degree     | 360 | +0.8     | [-0.6, +2.2]   | 0.4512   | +5.5   |
|                 | filler     | 369 | +1.4     | [+0.3, +2.7]   | 0.0597   | +9.0   |
|                 | all        | 369 | +1.4     | [+0.3, +2.7]   | 0.0597   | +9.0   |
| node_degree     | components | 395 | -1.0     | [-4.3, +2.3]   | 0.6543   | -1.0   |
|                 | clustering | 396 | +1.0     | [-2.8, +4.8]   | 0.7010   | +1.5   |
|                 | rwse       | 394 | +2.5     | [-1.5, +6.9]   | 0.2898   | +2.5   |
|                 | degree     | 376 | +11.7    | [+7.4, +16.0]  | 0.0001   | +8.8   |
|                 | filler     | 396 | -3.5     | [-7.3, +0.3]   | 0.0954   | -3.0   |
|                 | all        | 396 | +11.6    | [+7.3, +15.9]  | 0.0001   | +12.2  |

Table 15: qwen3-1.7b-think: all cells.

| Task            | Primer     | $n$ | $\Delta$ | 95% CI         | perm $p$ | zeroed |
|-----------------|------------|-----|----------|----------------|----------|--------|
| connected_nodes | components | 400 | +9.2     | [+6.0, +12.8]  | 0.0001   | +9.2   |
|                 | clustering | 400 | +1.0     | [-3.0, +4.8]   | 0.6893   | +1.0   |
|                 | rwse       | 400 | +0.0     | [-4.0, +4.0]   | 1.0000   | +0.0   |
|                 | degree     | 400 | -3.0     | [-7.2, +1.2]   | 0.1882   | -3.0   |
|                 | filler     | 400 | -12.5    | [-17.0, -8.0]  | 0.0001   | -12.5  |
|                 | all        | 400 | -2.8     | [-7.0, +1.5]   | 0.2511   | -2.8   |
| cycle_check     | components | 391 | -1.3     | [-2.6, +0.0]   | 0.1318   | -2.0   |
|                 | clustering | 395 | +0.3     | [+0.0, +0.8]   | 1.0000   | +0.5   |
|                 | rwse       | 392 | +0.3     | [+0.0, +0.8]   | 1.0000   | -0.2   |
|                 | degree     | 391 | -1.5     | [-2.8, -0.5]   | 0.0312   | -2.2   |
|                 | filler     | 392 | -0.3     | [-0.8, +0.0]   | 1.0000   | -0.5   |
|                 | all        | 395 | -0.5     | [-1.5, +0.5]   | 0.6269   | -0.2   |
| edge_count      | components | 397 | -1.3     | [-2.8, +0.3]   | 0.1756   | -1.2   |
|                 | clustering | 399 | -1.3     | [-3.0, +0.3]   | 0.2298   | -1.2   |
|                 | rwse       | 398 | -1.5     | [-3.0, -0.3]   | 0.0668   | -1.5   |
|                 | degree     | 393 | +27.7    | [+23.2, +32.6] | 0.0001   | +27.3  |
|                 | filler     | 400 | +1.2     | [-0.5, +3.2]   | 0.3012   | +1.2   |
|                 | all        | 397 | +3.8     | [+1.3, +6.3]   | 0.0049   | +3.8   |
| edge_existence  | components | 400 | -2.8     | [-6.0, +0.5]   | 0.1254   | -2.8   |
|                 | clustering | 400 | +4.0     | [+1.0, +7.0]   | 0.0168   | +4.0   |
|                 | rwse       | 400 | -10.0    | [-14.0, -6.2]  | 0.0001   | -10.0  |
|                 | degree     | 398 | -4.3     | [-7.8, -1.0]   | 0.0200   | -4.8   |
|                 | filler     | 400 | -7.0     | [-10.8, -3.5]  | 0.0002   | -7.0   |
|                 | all        | 399 | -9.8     | [-13.8, -6.0]  | 0.0001   | -10.0  |
| node_count      | components | 400 | +9.0     | [+6.2, +12.0]  | 0.0001   | +9.0   |
|                 | clustering | 400 | +8.8     | [+5.8, +11.8]  | 0.0001   | +8.8   |
|                 | rwse       | 400 | +9.2     | [+6.2, +12.2]  | 0.0001   | +9.2   |
|                 | degree     | 400 | +8.0     | [+5.0, +11.2]  | 0.0001   | +8.0   |
|                 | filler     | 400 | +9.2     | [+6.5, +12.2]  | 0.0001   | +9.2   |
|                 | all        | 400 | +9.8     | [+7.0, +12.8]  | 0.0001   | +9.8   |
| node_degree     | components | 400 | +0.2     | [+0.0, +0.8]   | 1.0000   | +0.2   |
|                 | clustering | 400 | -0.5     | [-2.0, +0.8]   | 0.7246   | -0.5   |
|                 | rwse       | 400 | -2.2     | [-4.0, -0.5]   | 0.0240   | -2.2   |
|                 | degree     | 400 | -6.5     | [-9.2, -3.8]   | 0.0001   | -6.5   |
|                 | filler     | 400 | -0.5     | [-2.0, +0.8]   | 0.7232   | -0.5   |
|                 | all        | 400 | -8.0     | [-10.8, -5.2]  | 0.0001   | -8.0   |

Table 16: qwen3-4b: all cells.

| Task            | Primer     | $n$ | $\Delta$ | 95% CI         | perm $p$ | zeroed |
|-----------------|------------|-----|----------|----------------|----------|--------|
| connected_nodes | components | 377 | +0.5     | [-0.5, +1.6]   | 0.6249   | +0.5   |
|                 | clustering | 373 | +0.0     | [-1.3, +1.3]   | 1.0000   | -0.8   |
|                 | rwse       | 379 | +0.0     | [-1.3, +1.6]   | 1.0000   | +0.5   |
|                 | degree     | 376 | +0.0     | [-1.6, +1.6]   | 1.0000   | +0.5   |
|                 | filler     | 379 | +0.8     | [-0.3, +2.1]   | 0.3787   | +1.2   |
|                 | all        | 375 | -0.5     | [-2.1, +0.8]   | 0.7306   | -1.2   |
| cycle_check     | components | 381 | +0.0     | [+0.0, +0.0]   | 1.0000   | -3.2   |
|                 | clustering | 394 | +0.0     | [+0.0, +0.0]   | 1.0000   | +0.0   |
|                 | rwse       | 395 | +0.0     | [+0.0, +0.0]   | 1.0000   | +0.2   |
|                 | degree     | 375 | -0.8     | [-1.9, +0.0]   | 0.2514   | -5.5   |
|                 | filler     | 396 | +0.0     | [+0.0, +0.0]   | 1.0000   | +0.5   |
|                 | all        | 392 | +0.0     | [+0.0, +0.0]   | 1.0000   | -0.5   |
| edge_count      | components | 60  | +1.7     | [+0.0, +5.0]   | 1.0000   | +24.8  |
|                 | clustering | 59  | +1.7     | [-3.4, +6.8]   | 1.0000   | +24.8  |
|                 | rwse       | 49  | +0.0     | [-6.1, +6.1]   | 1.0000   | +10.2  |
|                 | degree     | 34  | -23.5    | [-38.2, -8.8]  | 0.0073   | +8.5   |
|                 | filler     | 50  | -2.0     | [-6.0, +0.0]   | 1.0000   | +12.2  |
|                 | all        | 41  | -41.5    | [-56.1, -26.8] | 0.0002   | +1.8   |
| edge_existence  | components | 396 | +0.0     | [+0.0, +0.0]   | 1.0000   | -0.5   |
|                 | clustering | 396 | +0.0     | [+0.0, +0.0]   | 1.0000   | -0.8   |
|                 | rwse       | 393 | +0.0     | [+0.0, +0.0]   | 1.0000   | -1.2   |
|                 | degree     | 398 | +0.0     | [+0.0, +0.0]   | 1.0000   | +0.0   |
|                 | filler     | 399 | +0.0     | [+0.0, +0.0]   | 1.0000   | +0.0   |
|                 | all        | 394 | +0.0     | [+0.0, +0.0]   | 1.0000   | -1.0   |
| node_count      | components | 400 | +0.0     | [-1.0, +1.0]   | 1.0000   | +0.0   |
|                 | clustering | 399 | +0.5     | [+0.0, +1.3]   | 0.4936   | +0.2   |
|                 | rwse       | 399 | +0.5     | [+0.0, +1.3]   | 0.5005   | +0.2   |
|                 | degree     | 399 | +0.5     | [+0.0, +1.3]   | 0.5005   | +0.2   |
|                 | filler     | 400 | +0.5     | [+0.0, +1.2]   | 0.4870   | +0.5   |
|                 | all        | 400 | +0.5     | [+0.0, +1.2]   | 0.4870   | +0.5   |
| node_degree     | components | 380 | -0.3     | [-0.8, +0.0]   | 1.0000   | +0.2   |
|                 | clustering | 386 | -0.3     | [-1.0, +0.5]   | 1.0000   | +1.8   |
|                 | rwse       | 381 | -2.1     | [-3.7, -0.8]   | 0.0072   | -1.2   |
|                 | degree     | 388 | +0.0     | [-0.8, +0.8]   | 1.0000   | +2.5   |
|                 | filler     | 386 | +0.0     | [+0.0, +0.0]   | 1.0000   | +2.0   |
|                 | all        | 383 | -0.8     | [-1.8, +0.0]   | 0.2466   | +0.5   |

Table 17: qwen3-4b-think: all cells.